

Prac 06.1

For this homework we are not going to provide you with any code, you'll have to build the face detector on your own.

Step 0

Run the necessary imports.

```
import cv2
import math
import numpy as np
from matplotlib import pyplot as plt
plt.rcParams['figure.figsize'] = [15, 10]
```

Step 1

Load an image (any image that contains faces).

```
img = cv2.imread('TAFHeader.jpeg')
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
plt.imshow(img)

<matplotlib.image.AxesImage at 0x1f461649c90>
```



Step 2

Load the dlib face predictor.

```
import dlib
detector = dlib.get_frontal_face_detector()
```

Step 3

Run the predictor on your image.

```
gray = cv2.cvtColor(img, cv2.COLOR_RGB2GRAY)
faces = detector(gray, 1)
```

Step 4

Draw bounding boxes around the detected faces and plot the image. Use different colour for each face.

```
colors = [(255, 0, 0), (0, 255, 0), (0, 0, 255), (255, 255, 0), (0, 255, 255), (255, 0, 255)]

img_copy = img.copy()

for i, face in enumerate(faces):
    x1 = face.left()
    y1 = face.top()
    x2 = face.right()
    y2 = face.bottom()
    cv2.rectangle(img_copy, (x1, y1), (x2, y2), colors[i %
len(colors)], 2)

# Plot the result
plt.imshow(img_copy)
plt.axis('off')
plt.show()
```



Step 5 (optional)

Repeat the process with a different and more challenging image (more faces, smaller faces, people with glasses, hats, helmets, etc.). How does the detector perform? Is it robust?